

ORF309/MAT380

Poisson Processes II:

Supperposition and Thinning

(pp. 102-106)

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Agenda for today

- Sum of two Poisson RVs
- Superposition
- Thinning
- Examples

Recap

- **Definition:** A random process $\{N_t\}_{t \geq 0}$ is called a **Counting Process** iff
 - $N_0 = 0$
 - $N_t \in \{0, 1, 2, \dots\}$ for $t \geq 0$ (i.e., N_t is increasing in t)
 - N_t increases exactly by 1 each time t changes
- **Definition:** **Poisson Process** with rate μ , is a counting process $\{N_t\}_{t \geq 0}$ with the following **properties**
 - $N_t - N_s \sim N_{t-s} \sim \text{Pois}(\mu(t-s)), \forall t > s$ (**stationary** increments)
 - $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r \leq s} \quad \forall t > s$ (**independent** increments)

Sum of 2 Poisson RVs

Sum of two Poisson RVs (without Calculus)

- **Example:** You have: $X \sim \text{Pois}(\mu)$ and $Y \sim \text{Pois}(\nu)$ with $X \perp\!\!\!\perp Y$. What is the distribution of the random variable: $X + Y$?? that is: $P\{X + Y = k\} = ??$
- **Note:** You can use Calculus to solve this, and you should do it as exercise! *(using infinite sums)*
- We will use results from Poisson Processes and **avoid** lengthy Calculus computations.
- **Recall** the difference between a Poisson RV and a Poisson Proc:
 - **Poisson RV** has the distribution of Poisson Proc at a **fixed time t**
 - For a Poisson RV, there is no time involvement
 - **Poisson Proc** involves **many Poisson RVs**, and is evolving through time

Sum of two Poisson RVs (without Calculus)

- How can we **associate** the RV $X \sim \text{Pois}(\mu)$ with Poisson Processes?

Sum of two Poisson RVs (without Calculus)

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- Consider the Poisson Proc $\{N_t\}_{t \geq 0}$ **with rate 1** (i.e., 1 arrival per time unit, $N_1 \sim \text{Pois}(1)$)
 - N_t represents the number of arrivals in the interval of length t , for every $t > 0 \rightarrow N_t \sim \text{Pois}(t)$

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- Is it safe to say that $N_\mu \sim \text{Pois}(\mu)$??

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- Consider the Poisson Proc $\{N_t\}_{t \geq 0}$ **with rate 1** (i.e., 1 arrival per time unit, $N_t \sim \text{Pois}(1)$)
 - N_t represents the number of arrivals in the interval of length t , for every $t > 0 \rightarrow N_t \sim \text{Pois}(t)$
- How can we find a Poisson RV with rate μ ?
- Is it safe to say that $N_\mu \sim \text{Pois}(\mu)$?? $\rightarrow$ YES because:
 - N_μ is the number of arrivals by time μ (time is **fixed** to μ , here).
 - Since N_μ is an RV of the Poisson Proc $\{N_t\}_{t \geq 0}$ **with rate 1**, then **# of arrivals by time μ , is μ** ($N_\mu \sim \text{Pois}(\mu)$)
 - We must then have $N_\mu \sim \text{Pois}(\mu)$

Sum of two Poisson RVs (without Calculus)

- We can use the **same procedure for** Y and come up with the random variable $N_v \sim \text{Pois}(v)$
- **BUT** what would the major problem be ??

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- So we must look for another Poisson RV
- We know that $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r \leq s} \quad \forall s < t$

Sum of two Poisson RVs (without Calculus)

- We can use the **same procedure for** Y and come up with the random variable $N_\nu \sim \text{Pois}(\nu)$
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- We know that $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r \leq s} \quad \forall s < t$
- Hence if we consider the RV: $N_{\mu+\nu} - N_\mu \sim \text{Pois}(\nu)$ then $N_{\mu+\nu} - N_\mu \perp\!\!\!\perp N_\mu$

Sum of two Poisson RVs (without Calculus)

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- Hence if we consider the RV: $N_{\mu+\nu} - N_\mu \sim \text{Pois}(\nu)$ then $N_{\mu+\nu} - N_\mu \perp\!\!\!\perp N_\mu$
- So, we have **constructed**: $X, N_\mu \sim \text{Pois}(\mu)$ **and** $Y, N_{\mu+\nu} - N_\mu \sim \text{Pois}(\nu)$

-
- So, the **Joint distribution** of X , Y must be the **same** as that of N_μ , $N_{\mu+\nu} - N_\mu$
 - But $N_\mu + (N_{\mu+\nu} - N_\mu) = N_{\mu+\nu} \sim \text{Pois}(\mu + \nu)$
 - And hence: $X + Y \sim \text{Pois}(\mu + \nu)$ ■
 - **Remark:** We found the distribution of the **new random variable** $X + Y$ without doing any Calculus computations. We just used results we proved about Poisson Processes.

Superposition (or merging)

Superposition

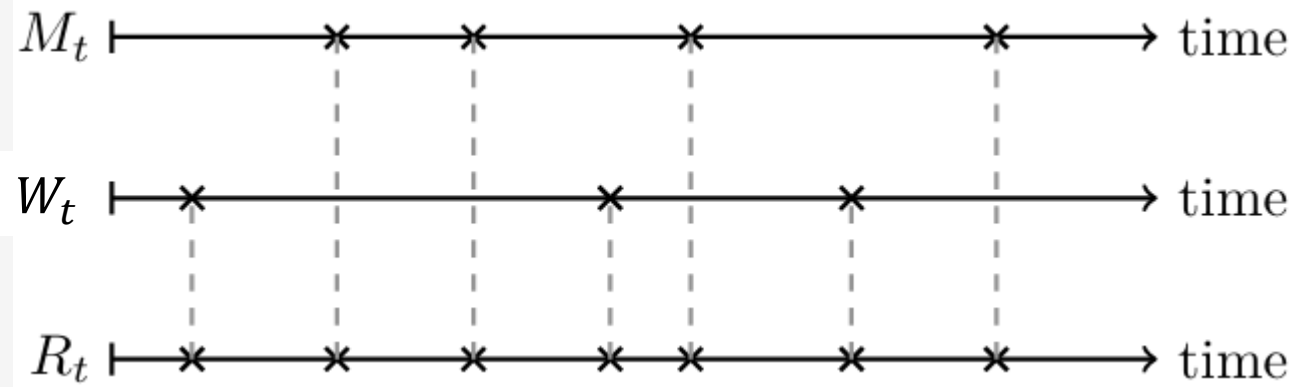
- Let's start with an **Example**:
 - On average ν women arrive at a store in 1 time unit
 - On average μ men arrive at the same store in 1 time unit
 - Men and Women arrivals are independent (no collusion)
- **Question**: what can we say about the process that counts the **total** number of customer arrivals ?

Superposition

- **Example (cont.):**
 - Let's define the random variables:
 - W_t = "# of women arriving at the store" $\sim \text{Pois}(\nu)$
 - M_t = "# of men arriving at the store" $\sim \text{Pois}(\mu)$
 - R_t = "total # of customers arriving at the store" $\sim ??$
 - R_t is a **new RV**, such that $R_t = W_t + M_t$

Superposition

- **Example (cont.):**

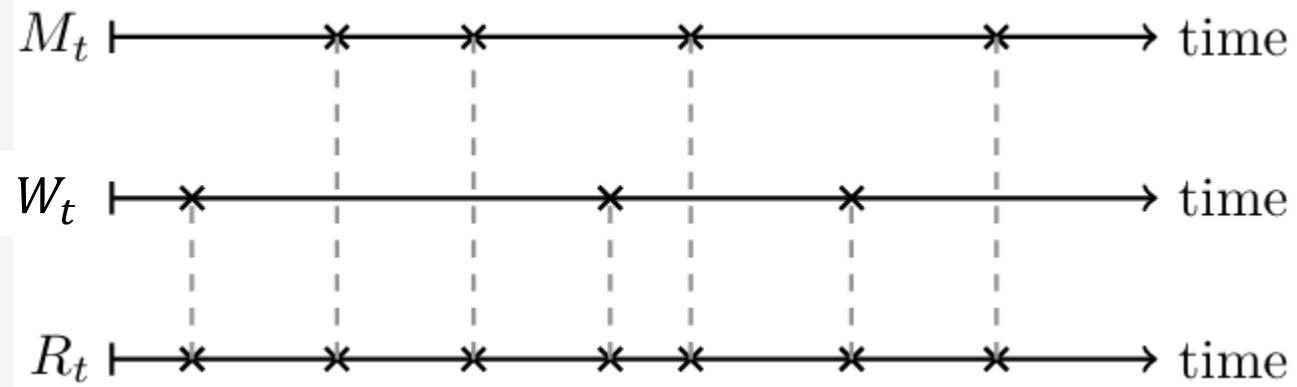


Observations for the Random Variables:

- R_t is obviously a Counting Process
- $R_t - R_s = (W_t - W_s) + (M_t - M_s)$
- $W_t - W_s \sim \text{Pois}(\nu(t - s))$ & $M_t - M_s \sim \text{Pois}(\mu(t - s))$
- $W_t - W_s \perp\!\!\!\perp M_t - M_s$
- **Hence:** $R_t - R_s \sim \text{Pois}((\nu + \mu)(t - s))$

Superposition

- **Example (cont.):**



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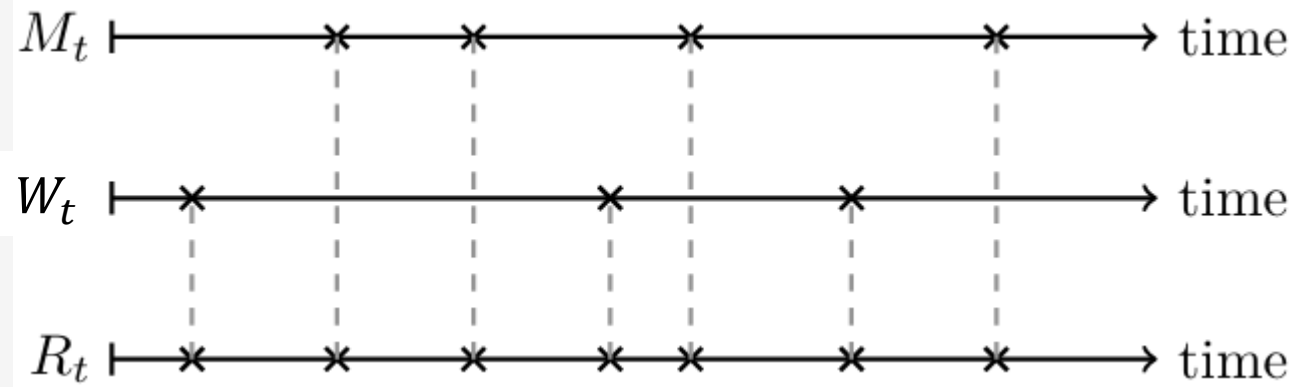
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- $W_t - W_s \perp\!\!\!\perp M_t - M_s$
- **Hence:** $R_t - R_s \sim \text{Pois}((\nu + \mu)(t - s))$

Observations for the Random Process

- $W_t - W_s \perp\!\!\!\perp \{W_r\}_{r \leq s}$ and $\{M_r\}_{r \leq s} \quad \forall s < t$
- $M_t - M_s \perp\!\!\!\perp \{M_r\}_{r \leq s}$ and $\{W_r\}_{r \leq s} \quad \forall s < t$
- **Hence:** $R_t - R_s \perp\!\!\!\perp \{R_r\}_{r \leq s} \quad \forall s < t$

Superposition

- **Example (cont.):**



Observations for the Random Variables:

- R_t is obviously a Counting Process
- $R_t - R_s = (W_t - W_s) + (M_t - M_s)$
- $W_t - W_s \sim \text{Pois}(\nu(t - s))$ & $M_t - M_s \sim \text{Pois}(\mu(t - s))$
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- **Hence:** $R_t - R_s \sim \text{Pois}((\nu + \mu)(t - s))$

Observations for the **Random Process**

- $W_t - W_s \perp\!\!\!\perp \{W_r\}_{r \leq s}$ and $\{M_r\}_{r \leq s} \quad \forall s < t$
- $M_t - M_s \perp\!\!\!\perp \{M_r\}_{r \leq s}$ and $\{W_r\}_{r \leq s} \quad \forall s < t$
- **Hence:** $R_t - R_s \perp\!\!\!\perp \{R_r\}_{r \leq s} \quad \forall s < t$

❖ So, we can conclude the $\{R_t\}_{t \geq 0}$ is a **Poisson Random Process with rate $\mu + \nu$**

Superposition

- **Principle of Superposition:**

- If we have k Poisson random processes $\{R_t^{(1)}\}_{t \geq 0}, \{R_t^{(2)}\}_{t \geq 0}, \dots, \{R_t^{(k)}\}_{t \geq 0}$ with rates $\mu_1, \mu_2, \dots, \mu_k$ and $R_t = R_t^{(1)} + R_t^{(2)} + \dots + R_t^{(k)}$ then $\{R_t\}_{t \geq 0}$ is a Poisson random process with rate $\mu = \mu_1 + \mu_2 + \dots + \mu_k$

Example in software engineering: We have 5 different types of bugs in a software system. The bug i appears as a Poisson Process with rate λ_i . What is the process of "total bugs found in the system?"

Thinning

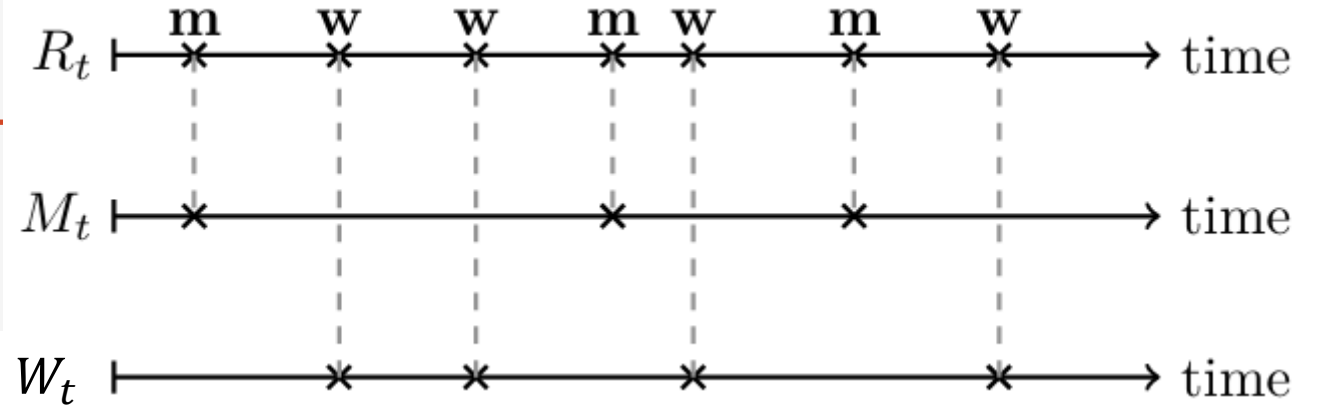
Thinning

- **Example:**
 - On average, μ customers arrive at a store per time unit.
 - A customer is independently a **Man** with probability p or a **Woman** with probability $1 - p = q$
 - **Question:** What can we say about the random processes that count the number of men and women, separately?

Thinning

- **Example (cont.):**

- **Random variables:**



- R_t = "# of customers arriving at the store by time t " $\sim \text{Pois}(\mu)$

- M_t = "# of Men customers arriving at the store by time t " $\sim ??$

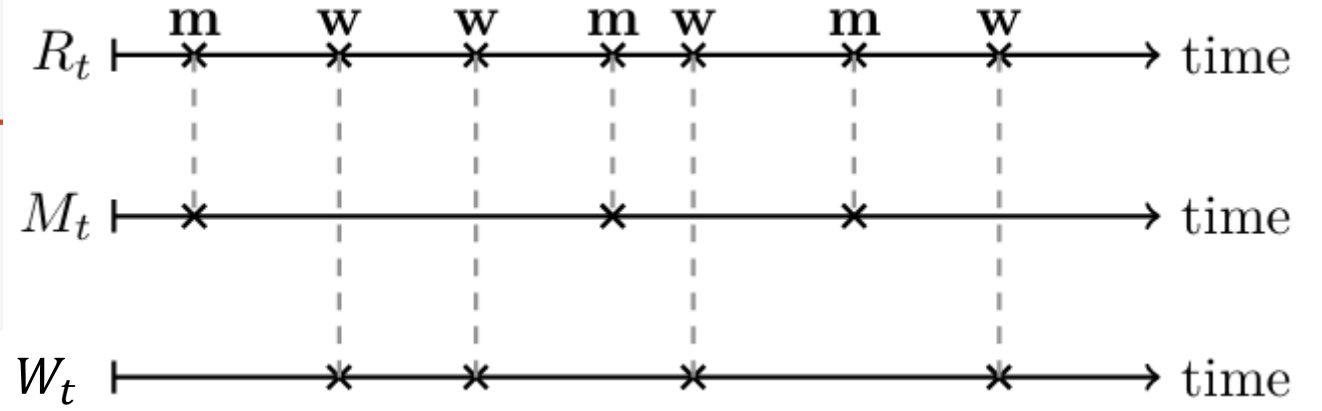
- W_t = "# of Women customers arriving at the store by time t " $\sim ??$

Thinning

- **Example (cont.):**

- **Random variables:**

- R_t = "# of customers arriving at the store by time t " $\sim \text{Pois}(\mu)$



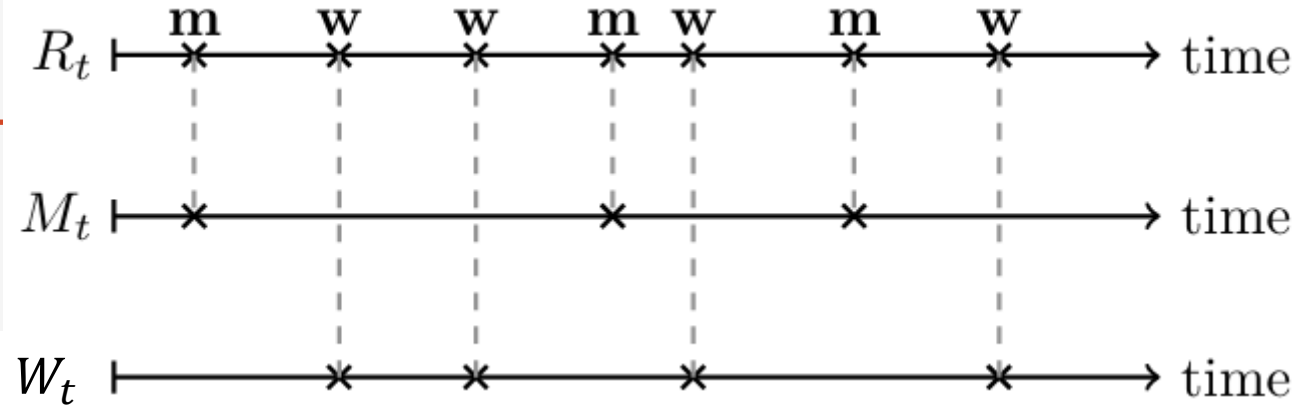
How can we use the information about the gender of the customers??

- M_t = "# of Men customers arriving at the store by time t " $\sim ??$
- W_t = "# of Women customers arriving at the store by time t " $\sim ??$

Thinning

- **Example (cont.):**

- **Random variables:**



- R_t = "# of customers arriving at the store by time t " $\sim \text{Pois}(\mu)$
- X_k = "the gender of the k -th customer" $\rightarrow P\{X_k = M\} = p$ and $P\{X_k = W\} = 1 - p$
- M_t = "# of Men customers arriving at the store by time t " $\sim ??$
- W_t = "# of Women customers arriving at the store by time t " $\sim ??$

$$M_t = \sum_{k=1}^{R_t} 1_{\{X_k=M\}} \quad W_t = \sum_{k=1}^{R_t} 1_{\{X_k=W\}}$$

Random variable

Diagram showing the decomposition of the total count R_t into the count of men M_t and the count of women W_t . The label "Random variable" has two green arrows pointing to the upper limits R_t of the summations in the equations above.

Thinning

- **Example (cont.):**

- Let's compute the **distribution of M_t**

Tower property

$$P\{M_t = m\} = \sum_{r=0}^{\infty} P\{M_t = m | R_t = r\} P\{R_t = r\} =$$

Thinning

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- Let's compute the **distribution of M_t**

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$$\begin{aligned} P\{M_t = m\} &= \sum_{r=0}^{\infty} P\{M_t = m | R_t = r\} P\{R_t = r\} = \\ &= \sum_{r=0}^{\infty} P\left\{ \sum_{k=1}^{R_t} 1_{\{X_k = M\}} = m | R_t = r \right\} P\{R_t = r\} = \end{aligned}$$

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- **Example (cont.):**

$$= \sum_{r=0}^{\infty} P \left\{ \sum_{k=1}^r 1_{\{X_k=M\}} = m \right\} P\{R_t = r\} =$$

Thinning

- **Example (cont.):**

$$\begin{aligned} &= \sum_{r=0}^{\infty} P \left\{ \sum_{k=1}^r 1_{\{X_k=M\}} = m \right\} P\{R_t = r\} = \\ &= \sum_{r=0}^{\infty} \frac{r!}{m! (r-m)!} p^m (1-p)^{r-m} \cdot \frac{e^{-\mu t} (\mu t)^r}{r!} = \end{aligned}$$

Thinning

- **Example (cont.):**

$$= \sum_{r=0}^{\infty} P \left\{ \sum_{k=1}^r 1_{\{X_k=M\}} = m \right\} P\{R_t = r\} =$$

$$= \sum_{r=0}^{\infty} \frac{r!}{m! (r-m)!} p^m (1-p)^{r-m} \cdot \frac{e^{-\mu t} (\mu t)^r}{r!} =$$

$$= \frac{p^m e^{-\mu t}}{m! (1-p)^m} \sum_{r=0}^{\infty} \frac{1}{(r-m)!} (1-p)^r \frac{(\mu t)^r}{1} = \frac{e^{-\mu t}}{m!} \frac{p^m}{(1-p)^m} \sum_{r=m}^{\infty} \frac{((1-p)\mu t)^r}{(r-m)!} =$$

Thinning

$$= \frac{e^{-\mu t}}{m!} p^m (1-p)^{-m} \sum_{r=m}^{\infty} \frac{((1-p)\mu t)^r}{(r-m)!} = \frac{e^{-\mu t}}{m!} \frac{p^m (\mu t)^m}{1} \sum_{r-m=0}^{\infty} \frac{((1-p)\mu t)^{r-m}}{(r-m)!} =$$

$\overbrace{e^{(1-p)\mu t}}$

Thinning

$$= \frac{e^{-\mu t}}{m!} p^m (1-p)^{-m} \sum_{r=m}^{\infty} \frac{((1-p)\mu t)^r}{(r-m)!} = \frac{e^{-\mu t}}{m!} \frac{p^m (\mu t)^m}{1} \sum_{r-m=0}^{\infty} \frac{((1-p)\mu t)^{r-m}}{(r-m)!} =$$

$$\frac{e^{-\mu t}}{m!} p^m (\mu t)^m e^{(1-p)\mu t} = \frac{e^{-\mu t}}{m!} p^m (\mu t)^m e^{\mu t} e^{-p\mu t}$$

Thinning

$$\begin{aligned} &= \frac{e^{-\mu t}}{m!} p^m (1-p)^{-m} \sum_{r=m}^{\infty} \frac{((1-p)\mu t)^r}{(r-m)!} = \frac{e^{-\mu t}}{m!} \frac{p^m (\mu t)^m}{1} \sum_{r-m=0}^{\infty} \frac{\overbrace{((1-p)\mu t)^{r-m}}^{e^{(1-p)\mu t}}}{(r-m)!} = \\ &\frac{e^{-\mu t}}{m!} p^m (\mu t)^m e^{(1-p)\mu t} = \frac{e^{-\mu t}}{m!} p^m (\mu t)^m e^{\mu t} e^{-p\mu t} \\ &= \frac{e^{-p\mu t} (p\mu t)^m}{m!} \end{aligned}$$

Hence: $M_t \sim \text{Pois}(p\mu t)$

Working similarly, we can show: $W_t \sim \text{Pois}((1-p)\mu t) \rightarrow \text{Exercise!!}$

Thinning

- It is easy to show that $\{M_t\}_{t \geq 0}$ and $\{W_t\}_{t \geq 0}$ have **stationary increments**
- **Hence** $\{M_t\}_{t \geq 0}$ and $\{W_t\}_{t \geq 0}$ are **Poisson processes** with rates $p\mu$ and $(1 - p)\mu$

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- **Hence** $\{M_t\}_{t \geq 0}$ and $\{W_t\}_{t \geq 0}$ are **Poisson processes** with rates $p\mu$ and $(1 - p)\mu$
- How about the **independence** of M_t and W_t ??

What do you think?

What does your intuition tell you?

Thinning

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- **Hence** $\{M_t\}_{t \geq 0}$ and $\{W_t\}_{t \geq 0}$ are **Poisson processes** with rates $p\mu$ and $(1 - p)\mu$
- How about the **independence** of M_t and W_t ??

$$P\{M_t = m, W_t = w\} = P\{M_t = m, R_t = m + w\} =$$

Thinning

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- How about the **independence** of M_t and W_t ??

$$P\{M_t = m, W_t = w\} = P\{M_t = m, R_t = m + w\} =$$

$$= P\left\{\sum_{k=1}^{R_t} 1_{\{X_k=M\}} = m, R_t = m + w\right\} = P\left\{\sum_{k=1}^{m+w} 1_{\{X_k=M\}} = m, R_t = m + w\right\} \quad \square$$

Thinning

$$= P \left\{ \sum_{k=1}^{m+w} 1_{\{X_k=M\}} = m \right\} P\{R_t = m + w\} = \binom{m+w}{m} p^m (1-p)^w \frac{e^{-\mu t} (\mu t)^{m+w}}{(m+w)!} =$$

Thinning

$$\begin{aligned} &= P \left\{ \sum_{k=1}^{m+w} 1_{\{X_k=M\}} = m \right\} P\{R_t = m + w\} = \binom{m+w}{m} p^m (1-p)^w \frac{e^{-\mu t} (\mu t)^{m+w}}{(m+w)!} = \\ &= \frac{(m+w)!}{m! w!} p^m (1-p)^w \frac{e^{-\mu t} (\mu t)^{m+w}}{(m+w)!} = \frac{p^m (1-p)^w}{m! w!} e^{-\mu t} (\mu t)^m (\mu t)^w = \end{aligned}$$

Thinning

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$$= \frac{1}{m! w!} e^{-\mu t} (p\mu t)^m ((1-p)\mu t)^w = \frac{1}{m! w!} e^{-\mu t + p\mu t - p\mu t} (p\mu t)^m ((1-p)\mu t)^w =$$

Thinning

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Thinning

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$$\underbrace{\frac{e^{-p\mu t} (p\mu t)^m}{m!}}_{P\{M_t = m\}} \cdot \underbrace{\frac{e^{-(1-p)\mu t} ((1-p)\mu t)^w}{w!}}_{P\{W_t = w\}}$$

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$$= \frac{1}{m! w!} e^{-\mu t} (p\mu t)^m ((1-p)\mu t)^w = \frac{1}{m! w!} e^{-\mu t + p\mu t - p\mu t} (p\mu t)^m ((1-p)\mu t)^w =$$

$$\underbrace{\frac{e^{-p\mu t} (p\mu t)^m}{m!}}_{P\{M_t = m\}} \cdot \underbrace{\frac{e^{-(1-p)\mu t} ((1-p)\mu t)^w}{w!}}_{P\{W_t = w\}} \Rightarrow P\{M_t = m, W_t = w\} = P\{M_t = m\} P\{W_t = w\}$$

Hence: $M_t \perp\!\!\!\perp W_t$

So, why are M_t and W_t independent??

- **The paradox:** We know $R_t = M_t + W_t$, and intuitively when we know one part (e.g., W_t) we should deduce the other (e.g., M_t)
- **But in Poisson Procs**, R_t is **not known**, it is a **random variable** (it is **not fixed!**)
- We only know μ , the average number of customer arrivals (men and women), say $\mu = 6$ per hour.
- If we observe $M_t = 8$, then there are two competing thoughts:
 - **The “subtraction thought”:** if R_t happened to be 10, then there must be $W_t = 2$
 - **The “rate thought”:** I was expecting $\mu = 6$ customers (men and women), but I already have observed 8 men. It must be a busy day for men. It reasonable to think that it must also expect more women!
 - **Result:** in Poisson Procs, these two thoughts **cancel each other**, resulting in M_t, W_t being **independent!**

Thinning

- **Principle of thinning**

- **If** $\{R_t\}_{t \geq 0}$ is a Poisson process with rate μ ,
- and each arrival is assigned a type in the set $\{1, 2, \dots, k\}$ with probabilities $p_1, p_2, \dots, p_k$ and $p_1 + p_2 + \dots + p_k = 1$,
- and $N_t^{(i)}$ = "# arrivals of type i by time t " for $i = 1, 2, \dots, k$,
- **Then** $\{N_t^{(i)}\}_{t \geq 0}$ for $i = 1, 2, \dots, k$ are independent Poisson processes with rates $p_i \mu$
